

TOOL TO SUPPORT GROUP MULTI CRITERIA DECISION-MAKING METHOD FOR R&D PROJECTS

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Abstract – In the new product Research and Development Projects (R&D), the decision-making process of the research crew to solve the newly emerged problem is a complex process because of including many stakeholders. Research crew, user, investor,... are hard to have mutual agreement due to the conflict in several aspects like expectation, demand, cognitive,... This is the source of the uncertainty of the decision-making process. This paper suggests a tool helping the decision-making process base on the multi-criteria method with multi-stakeholder in which the consensus progress is promoted by the manager and tools, through identifying conflicting points and following their evolution after several rounds of discussion. The tool is expected to shorten decision-making time and provide feedback as suggestions to help those who need to modify their preferences during discussion to reach a consensus.

Key words - Research and Development (R&D); Multi-Criteria; Group Decision-Making; Consensus.

Tóm tắt – Ở các dự án Nghiên cứu và Phát triển (R&D) những sản phẩm mới, việc nhóm nghiên cứu đưa ra quyết định để giải quyết các vấn đề nảy sinh là một quá trình phức tạp vì nó bao gồm nhiều bên liên quan. Nhóm nghiên cứu, người dùng, nhà đầu tư,... thường khó đạt được sự đồng thuận vì mâu thuẫn ở những yếu tố như: kì vọng, nhu cầu, nhận thức,... đây là nguồn gốc của sự không chắc chắn trong quá trình ra quyết định. Bài báo này đề xuất một công cụ hỗ trợ việc ra quyết định dựa trên phương pháp đa tiêu chí với nhiều bên liên quan, trong đó quá trình đồng thuận được thúc đẩy bởi người điều hành và công cụ hỗ trợ, thông qua xác định các điểm mâu thuẫn và theo dõi sự thay đổi của chúng qua từng vòng thảo luận. Công cụ được kì vọng sẽ rút ngắn thời gian khi ra quyết định và cung cấp phản hồi là các đề xuất để giúp những người cần sửa đổi sở thích của họ trong quá trình thảo luận để đạt được sự đồng thuận chung.

Từ khóa - Nghiên cứu và phát triển (R&D); Đa tiêu chí; Quyết định nhóm; Đồng thuận.

1. Problem Identification

In research and development (R&D) projects, arising issues will always directly affect the interests of the parties involved. Therefore, the decision-making process to solve these problems is often complicated, and lengthy and affects the progress and quality of the project.

1.1. Research and Development (R&D)

R&D is working to improve or develop products, services, technologies, etc., or even a theoretical foundation that can create value to bring profits to businesses, organizations, and society [1]. Therefore, R&D is present in almost every field of society such as computer science, materials science, health science,... R&D can be carried out by several individuals, research groups, and organizations in a variety of contexts such as in specialized laboratories, companies, and universities. According to the collected literature, R&D can be divided into two groups: R&D of products with tangible value (R&D of new products, services, ...) and intangible value (new theory, additional testing, old theory,...) [2].

1.2. Decision-making in R&D projects

This model is based on the Multi-Criteria Decision-Making method (MCDM), which includes the following steps: Identify the problem to be solved, analyze the stakeholders, identify the solutions, determine the criteria, evaluate the weight of the criteria, and finally, the evaluation of the stakeholders come up with a solution [3].

1.3. Urgency of the subject

Currently, there have been a few research topics on tools to support decision-making, but the application level is not widespread and automation still has many shortcomings [4]. Especially, the field of R&D is still new, so few people are interested. The tool proposed by this paper is expected to shorten decision-making time and

provide feedback to support decision-making to optimize consensus. Besides, it also overcomes the weaknesses of existing tools such as discussion time, discussion issues, and user experience.

2. Research objectives

The tool called "DeMas" is programmed by the main programming language Python, which is a web app. Combined with some current web technologies such as Ajax, "DeMas" is also a Django project, so response time and data processing are improved.

The method proposed in this paper is adapted from the Multi-criteria decision-making method-MCDM and allows stakeholders to discuss together to reach a consensus on the choice of solution. In this method, the AHP (Analytical Hierarchy Process) method is used in building the hierarchical structure of the problem and determining the weight of the criteria used in the evaluation [5].

This model consists of 7 steps, respectively [7]:

2.1. Step 1 – Identify the problem

It includes many small steps such as: defining the problem, defining the solutions, and classifying the solutions. This is the premise for the group decision-making process, of the stakeholders involved in the discussion.

2.2. Step 2 – Stakeholder Analysis

Consists of three sub-steps including identifying stakeholders, specifying their relationships, and the weights of each of them. Basic factors such as professional knowledge, experience, and skills, ... are the criteria used to determine the weight of each individual in a group.

2.3. Step 3 – Identify the alternatives

Multi-criteria decision-making (MCDM) is a way of handling decisions regarding the choice of the best solution among potential solutions. Depending on a number of qualitative or quantitative criteria or attributes, proposed solutions can take many forms and are customized to specific situations and problems.

2.4. Step 4 –Identify the Criteria

Decision-making entails comparing different solutions by analyzing the attributes that the solutions have to determine the importance, compare and obtain a relative choice concerning the objectivity of each attribute, and then aggregate the results to get a solution with high consensus.

As mentioned, in this article, I decided to choose the AHP method, which is a relatively simple and systematic approach, used by those involved in the decision-making process.

2.5. Step 5 - Weighting of criteria

Stakeholders rate the relative importance of a criterion relative to each other. Each participant gives his or her opinion reflecting the level of preference for each pair of criteria.

Based on the rating of each participant, the weight of the criteria will be calculated. This means that each participant has a set of criteria for evaluating solutions with different priorities.

2.6. Step 6 – Consensus Finding

Traditionally, the weight of the criteria is ranked, without taking into account the degree of agreement among the stakeholders. To approach consensus, an iterative process for stakeholders to express, discuss, and revise their assessment to reach consensus before a joint decision is made.

The term consensus can be defined as the state in which a decision is made to the satisfaction of the majority of the group members [6]. Reaching a consensus means that the participants change their initial opinion in the discussion, towards the decision with the highest consensus. If consensus cannot be reached, a feedback proposal is sent to the stakeholders to modify their preferences to increase the level of agreement in subsequent rounds of discussions. This is the novelty of the method proposed by this paper.

The level of agreement among the participants is calculated according to the following 4 steps:

2.6.1. Normalization

2.6.2. Calculate Consensus

2.6.3. Consensus Control

2.6.4. Feedback

2.7. Step 7 – Proposed Solutions

This step represents the final aggregation of the weights obtained and the evaluation of the alternatives on each criterion. After the consensus process, different points were considered for compromise, and the most consensual decision was made.

Based on the weighting of the criteria and the evaluation of alternatives, each participant will have a different rating for the alternatives.

3. Research Results

3.1. Overview

In this paper, the tool is applied to evaluate the production option of technological life jackets - sCoat through the defined criteria [8]. Technological life jackets are considered a new product to serve Vietnamese fishermen. The process of forming and perfecting sCoat has the same process as the process of R&D of new products through the program "Engineering Projects in Community Service" - (EPICS) [9].

To choose the right production method, many criteria need to be evaluated by the stakeholders (Fishers, implementation team,...) and the tools that support this process.

3.2. Perform

The survey team discussed together choosing the sCoat production option through the support tool.

The screenshot shows a web-based questionnaire titled "Which is better?". It includes fields for "Question", "Alternative", and "Criteria". Below these is a "Submit" button. A comparison matrix is displayed with the following data:

	1. sCoat Mini Kit	2. sCoat Original	3. sCoat Pro	4. sCoat Pro Max
Time	1	0.8	0.14	0.1
Risk	0.8	1	0.67	0.5
Price	1	0.8	0.5	0.3
Production	1	1	0.3	0.3
Fashion	0.56	0.67	0.89	1

Figure 1: User Interface Section

4. Discussion

4.1. Review

After the first round of the discussion round ended, here are the results of the user, investor, and research team's evaluation, respectively.

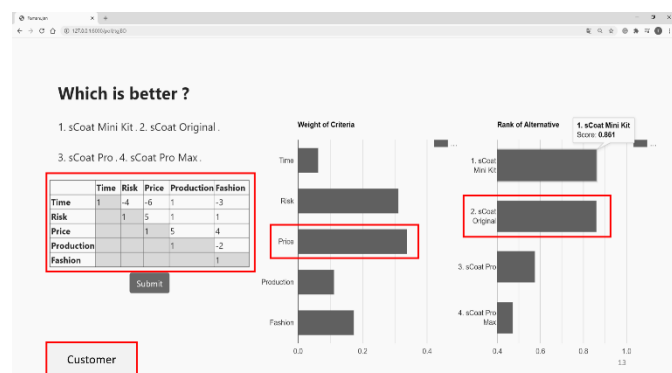


Figure 2: Results returned after customer evaluation

It can be seen that the role of the user is concerned about the price of the product on the opposite side, the investor focuses on production ability more than the other criteria.



Figure 3: Results returned after investor evaluation

About research crew, they focus on solving risks while using the new product.

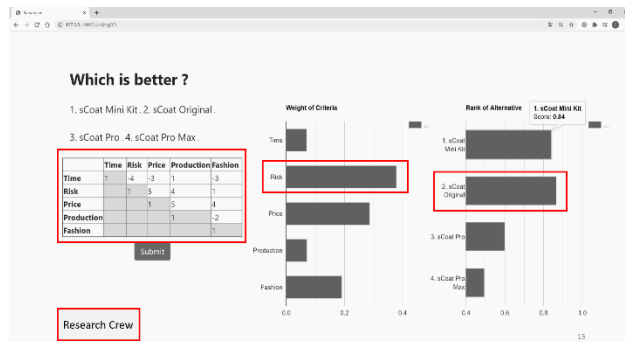


Figure 4: Results returned after research crew evaluation

At the end of the first round, the results showed that investors were in conflict with the rest of the group.

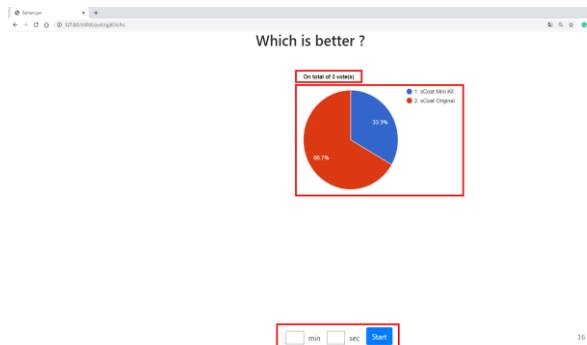


Figure 5: Results of the first round of discussions

The tool discovered that inconsistency, and in the next round of discussions, the whole group would discuss the manufacturability criteria.

After many rounds of discussion, when investors were less worried about the product's production ability, the choice gradually shifted from Mini Kit to Original.

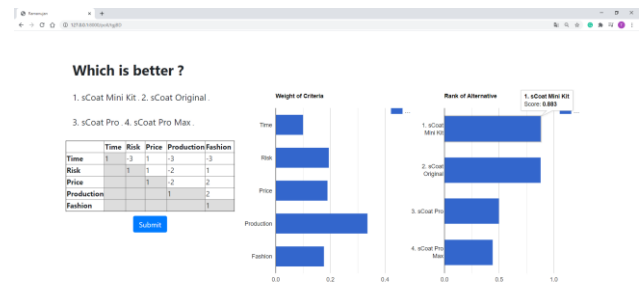


Figure 6: Investor evaluation results after the discussion round

Through the tool after many rounds of discussion, the results of the discussion conducted by the project team have reached a consensus on the production plan of sCoat Original.

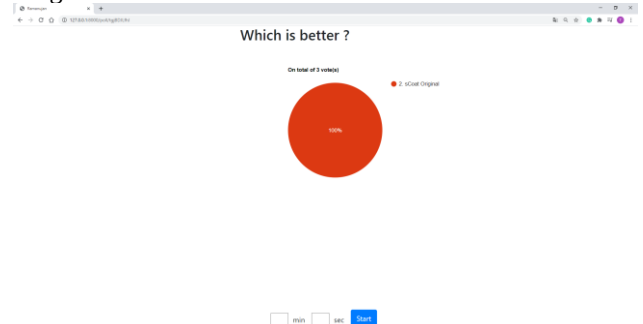


Figure 7: Results of the final round of discussions

5. Conclusion

Currently, the tool proposed by this paper has shortened the decision-making time and provided feedback to support decision-making to optimize consensus. In addition, the tool also has some interface and user experience problems when applied in large-scale models.

Web App: <http://demas.herokuapp.com>

In the future, the tool is expected to provide more "smart" responses to help the discussion process become more efficient and save time.

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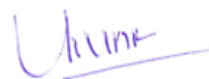
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